Softmax Regression:
A Multinoulli Whose Parameter Is a Function of X

Yuval Lavie

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Abstract

Bernoulli → multinoulli: for K classes, the conditional label distribution is a multinoulli whose probability vector depends on x through the softmax of K linear scores. Conditional MLE is cross-entropy minimization; the gradient is (softmax − one-hot) ⊗ features — residual times features, vectorized over classes; $K = 2$ collapses to logistic regression. As there, no closed form, convexity, and a Bayes floor $\mathbb{E}[1 - \max_k p_k(X)]$ that the well-specified fit approaches. All demonstrated in `softmax.py`.

1 The conditional-multinoulli view

For $Y \in \{1, \dots, K\}$, whatever the joint over (X, Y) is,

$$Y \mid X = x \;\sim\; \text{Multinoulli}\big(p_1(x), \dots, p_K(x)\big), \qquad p_k(x) = \Pr(Y = k \mid X = x), \tag{1}$$

automatically — a distribution on K outcomes *is* a multinoulli; the modeling assumption is again only the form of the parameter *function*, now simplex-valued. The linear model gives every class a score $w_k^\top x$ (rows of $W \in \mathbb{R}^{K \times d}$, x carrying the leading 1) and maps scores to the simplex by softmax:

$$p_k(x) \;=\; \frac{e^{w_k^\top x}}{\sum_{j=1}^K e^{w_j^\top x}} \;=\; \text{softmax}(Wx)_k, \tag{2}$$

softmax being to the simplex what the sigmoid is to $(0, 1)$ (distributions note). Log-odds between any two classes are linear: $\log\big(p_k(x)/p_j(x)\big) = (w_k - w_j)^\top x$, so all pairwise decision boundaries are hyperplanes.

Remark 1 (Identifiability)**.** *Adding one fixed vector v to every row of W leaves* (2) *unchanged ($e^{v^\top x}$ cancels), so W is only identified up to a shared row-shift (conventionally fixed by pinning $w_K = 0$). The identifiable object — and what the experiment evaluates — is the probability function* (2) *itself.*

2 Conditional MLE = cross-entropy

The conditional likelihood of a sample and its average NLL:

$$\mathcal{L}(W) \;=\; -\frac{1}{n}\sum_{i=1}^n \log p_{y_i}(x_i) \;=\; \frac{1}{n}\sum_{i=1}^n \Big[\log \sum_j e^{w_j^\top x_i} \;-\; w_{y_i}^\top x_i\Big], \tag{3}$$

the multiclass cross-entropy (the log-sum-exp playing the role of the Bernoulli note's $\log(1+e^t)$ — it is the log-normalizer $A(\theta)$ of the multinoulli exponential family).

Proposition 1 (Gradient)**.** *With* $p^{(i)} = \mathrm{softmax}(Wx_i)$ *and one-hot* e_{y_i}*,*

$$\boxed{\nabla_{w_k}\mathcal{L}_i \;=\; \big(p_k(x_i) - \mathbf{1}[y_i = k]\big)\, x_i, \qquad \nabla_W \mathcal{L}_i = \big(p^{(i)} - e_{y_i}\big)\, x_i^\top.} \tag{4}$$

Proof. Differentiate (3) in w_k: $\partial_{w_k} \log \sum_j e^{w_j^\top x} = \frac{e^{w_k^\top x}}{\sum_j e^{w_j^\top x}}\, x = p_k(x)\, x$, and $\partial_{w_k} w_y^\top x = \mathbf{1}[y = k]\, x$. Subtract. □

Residual times features once more — the GLM identity of the Poisson note, with the multinoulli as the vector-valued exponential family. For $K = 2$, subtracting the two rows of (4) recovers the logistic gradient $(\sigma - y)x$ exactly. Stationarity $\sum_i p^{(i)} x_i^\top = \sum_i e_{y_i} x_i^\top$ is transcendental (no closed form), the Hessian is PSD (log-sum-exp is convex; composed with the linear map), so gradient methods own the problem — and the separable-data pathology of the logistic note recurs identically: on perfectly separated classes the MLE does not exist.

3 The multiclass Bayes floor

For any classifier h,

$$\Pr\big(h(X) \neq Y\big) \;\geq\; \mathbb{E}\Big[1 - \max_k\, p_k(X)\Big], \tag{5}$$

attained by $h^*(x) = \arg\max_k p_k(x)$: at each x, predicting the likeliest class errs with probability $1 - \max_k p_k(x)$ and nothing does better. The binary floor $\mathbb{E}[\min(p, 1-p)]$ is the $K = 2$ case.

4 Experiment

$K = 3$ classes in $\mathbb{R}^2$, $n = 6000$ labels drawn from a true softmax model (seed 7). GD converges in the single convex basin; hand and autograd (`F.cross_entropy`) SGD trajectories coincide to 10^{-10}; fitted class probabilities match the true ones to $\approx 10^{-2}$ mean absolute error (the identifiable comparison — not W vs W^*); and the fitted classifier's test error meets the Bayes floor (5) to three decimals. The figure shows the three linear decision regions, the convex convergence, and the calibration diagonal for $p_0(x)$ — the estimated object is, once again, a *parameter function*, now simplex-valued.